

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. [Review this update](#) and manage your preferences in your [GitHub account settings](#). 

 **DIPT-EXP-2-Record-Image-Acquisition-using-Web-Camera**
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EX. NO:02

IMAGE ACQUISITION USING WEB CAMERA

Aim

To write a Python program using OpenCV to capture an image from the webcam and perform the following operations:

1. Write the frame as a JPG file
2. Display the video
3. Display the video by resizing the window
4. Rotate and display the video

Software Used

- Anaconda – Python 3.7
- Jupyter Notebook / VS Code
- OpenCV (`cv2`)

Algorithm

Step 1:

Import the required libraries and initialize the webcam using `cv2.VideoCapture()` .

Step 2:

Capture frames continuously from the webcam.

Step 3:

Save a frame as a JPG image using `cv2.imwrite()` .

Step 4:

Display the live video stream using `cv2.imshow()` .

Step 5:

Resize the frame and rotate it using OpenCV functions, then display the processed frames.



Program

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```
import cv2
import matplotlib.pyplot as plt
from IPython.display import clear_output
import time
cap = cv2.VideoCapture(0)
ret, frame = cap.read()
if ret:
    cv2.imwrite("captured_frame.jpg", frame)
cap.release()
captured_image = cv2.imread('captured_frame.jpg')
plt.imshow(captured_image[:, :, :-1])
plt.title('Captured Frame')
plt.axis('off')
plt.show()
cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()
    if not ret:
        break
    frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    clear_output(wait=True)
    plt.imshow(frame_rgb)
    plt.axis('off')
    plt.show()
    time.sleep(0.05)
cap.release()
cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()
    if not ret:
        break
    resized_frame = cv2.resize(frame, (100, 150)) # Resize to 320x240
    frame_rgb = cv2.cvtColor(resized_frame, cv2.COLOR_BGR2RGB)
    clear_output(wait=True)
    plt.imshow(frame_rgb)
```



```
plt.axis('off')
plt.show()
time.sleep(0.05)
cap.release()

cap = cv2.VideoCapture(0)

for i in range(50):
    ret, frame = cap.read()
    if not ret:
        break
    rotated_frame = cv2.rotate(frame, cv2.ROTATE_90_CLOCKWISE)
    frame_rgb = cv2.cvtColor(rotated_frame, cv2.COLOR_BGR2RGB)
    clear_output(wait=True)
    plt.imshow(frame_rgb)
    plt.axis('off')
    plt.show()
    time.sleep(0.05)
cap.release()
```

Output

1. CAPTURED FRAME



2. DISPLAYING THE VIDEO



3. DISPLAYING THE VIDEO BY RESIZING THE WINDOW



4. ROTATE AND DISPLAY THE VIDEO



Result

Thus, the image is successfully captured from the webcam and various video processing operations such as saving, displaying, resizing, and rotating are performed using OpenCV.